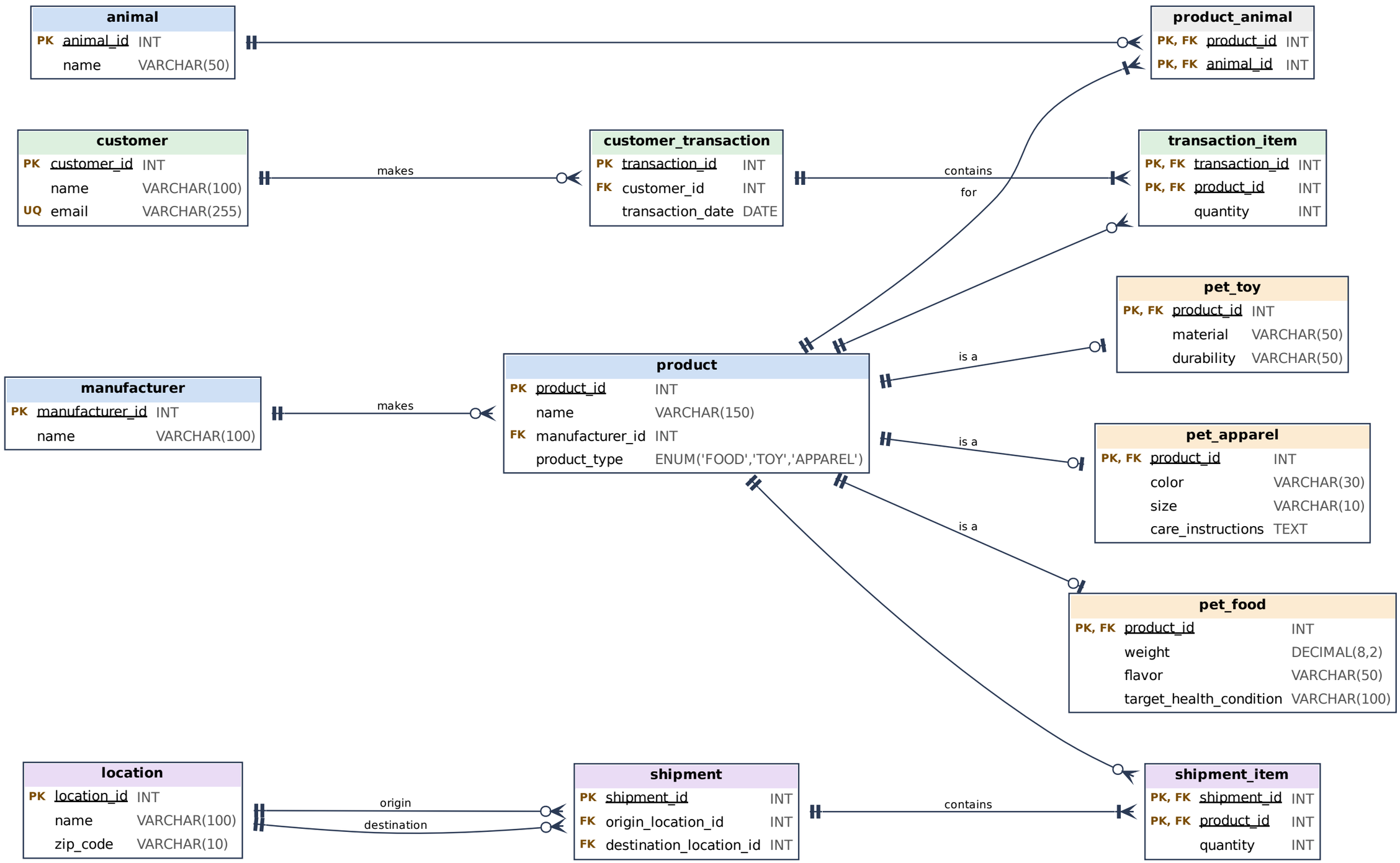
Walmart Pet Department: Centralized Database ERD

Normalized to 3NF | Crow's foot notation

Legend (crow's foot)	
-||	exactly one
-o|	zero or one
-|<	one or many
-o<	zero or many
PK / FK / UQ	primary / foreign / unique key
	product catalog
	product subtypes
	customers and sales
	locations and shipments
	junction table



Design Notes: Pet Department Database

Requirements coverage

Requirement	Where it lives in the schema
Food, toy and apparel products with their attributes	**product** holds shared columns (name, manufacturer); **pet_food**, **pet_toy**, **pet_apparel** hold type-specific columns
Each product has one or more animals	**product_animal** junction table; the product side is one-or-many
Each product has a manufacturer	**product.manufacturer_id** is a NOT NULL foreign key to **manufacturer**
Customer names and emails	**customer**, with email marked UNIQUE
Transactions with a date and one or more products	**customer_transaction** (date, customer) plus **transaction_item** (one row per product)
Locations with a name and zip code	**location**
Shipments: origin, destination, products with quantities	**shipment** has two foreign keys to **location**; **shipment_item** stores product and quantity

Key design decisions

Supertype/subtype for products. Name and manufacturer are common to every product, so they live once in **product**. Each subtype table shares the product's primary key (a 1-to-0..1 relationship), and **product_type** says which subtype applies. This lets transactions and shipments reference a single **product** table instead of three, and avoids a wide table full of NULLs such as a toy with a flavor column.
Junction tables for many-to-many relationships. Products and animals, transactions and products, and shipments and products are all many-to-many, so each gets a junction table with a composite primary key. The composite key also stops the same product from being listed twice in one transaction or shipment.
Lookup tables for repeated values. Manufacturer and animal are separate tables rather than free text, so a manufacturer's name is stored once and can be renamed in one place.
Two foreign keys from shipment to location. Origin and destination are both locations, so **shipment** references **location** twice in different roles.

Normalization (3NF)

1NF: every column holds a single value; multi-valued facts (animals per product, products per transaction or shipment) are rows in junction tables, never lists. **2NF:** in tables with composite keys, quantity depends on the whole key, not just one part. **3NF:** no non-key column depends on another non-key column; for example, a manufacturer's details are not repeated on each product.

Assumptions

quantity on transaction_item is an addition beyond the brief, since a customer can buy more than one of the same product. The transaction table is named **customer_transaction** because TRANSACTION is a reserved word in SQL. Weight, durability, size and target health condition are kept as single attributes; if the department needs to filter heavily on health conditions, that column could become its own lookup table in a later revision.